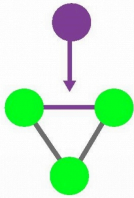
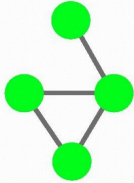


L4: Link Selection Model

a. NEW NODE



b.



Another very simple generating model that also creates power-law networks with exponent 3 is the **“link selection”** model.

Suppose that each time we introduce a new node, we select a random link and the new node connects to one of the two end-points of that link (*randomly chosen*). In other words, the new node connects to a randomly selected edge-stub (*see visualization*).

In this model, the probability that the new node connects to a node of degree k is proportional to k (*because the node of degree- k has k stubs*). But this is exactly the same condition with the preferential attachment model: a linear relation between the degree k of an existing node and the probability that the new node connects to that existing node of degree- k .

So, the link selection model is just a variant of preferential attachment and it also produces power-law degree distribution with exponent $\alpha=3$.